

# CPE 486/586: Deep Learning for Engineering Applications

06 Generative Adversarial Network

Spring 2026

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# Outline

1. Motivation
2. Generative Adversarial Networks
3. GANs (in PyTorch)
4. Examples
5. Challenges with GANs
6. Advance Architecture in GAN

# Lecture Overview

- ⚡ In this lecture we discussed generative models (briefly) and how to model distributions of data using density estimation techniques. One advantage to these generative models is that we can generate data from the distribution once we have the model.
- ⚡ Our goal in this lecture is to introduce *Generative Adversarial Networks* (GANs) for generating data from a distribution  $p_{data}(\mathbf{x})$ . GANs are able to model complex distributions of data that would be incredible challenging to model using the density estimation approaches discussed in previous lectures.
- ⚡ Modeling  $p_{data}(\mathbf{x})$  is increasingly challenging for complex data, such as real scenes, people, etc.

# Motivation

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# Discriminative vs. Generative Models

## Discriminative

Focuses on the **boundary** between classes.

- ⚡ Maps high-dimensional input  $\mathbf{x}$  to label  $y$ .
- ⚡ Learns  $P(y | \mathbf{x})$ .
- ⚡ Goal: Classification and Regression.

## Generative

Focuses on the **distribution** of the data.

- ⚡ Models the process that *creates* the data.
- ⚡ Learns  $P(\mathbf{x}, y)$  or  $P(\mathbf{x})$ .
- ⚡ Strategy: MLE or Variational Inference.

While **Discriminative** models tell you how to tell things apart, **Generative** models give you the "recipe" for the data itself.

**GAN is a generative model.**

# Adversarial Machine Learning

## Wikipedia Definition

Study of attacks on ML algorithms and defenses against such attacks.

- ⚡ **Data Assumption:** Conventional training is based on IID assumption, but however someone can supply non-IID data.
- ⚡ **Exploitation:** Adversaries may attempt to exploit a learning mechanism either to cause it to misbehave or to extract or misuse information.



While we won't delve in Secure Machine Learning, we serve this as a motivation for **Generative Adversarial Network (GAN)**.

GAN is a **two-player minimax** game where:

**Generator:** A counterfeiter (adversary) trying to fool you.

**Discriminator:** A police officer investigating and detecting counterfeit objects.

## Example of a Complex $p_{data}(\mathbf{x})$



## Example of a Complex $p_{data}(\mathbf{x})$



## Example of a Complex $p_{data}(\mathbf{x})$



## Example of a Complex $p_{data}(\mathbf{x})$



None of these people are real!



There are hundreds(?) of different types of GANs; however, this lecture only presents the vanilla GAN.

# Generative Adversarial Networks

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# Definitions

## Data Distribution

$p_{data}(\mathbf{x})$ : Probability distribution over the data.

⚡ Access to  $\{\mathbf{x}_i\}_{i=1}^n$

⚡ No labels needed (not classification).

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## Generator $G(\mathbf{z})$

$G$  is the generator neural network that takes in a noise vector,  $\mathbf{z}$ , to produce a vector  $\mathbf{x}_{fake}$ .  $\mathbf{x}_{fake}$  is a “fake” data sample that is generated from  $\mathbf{z}$ .

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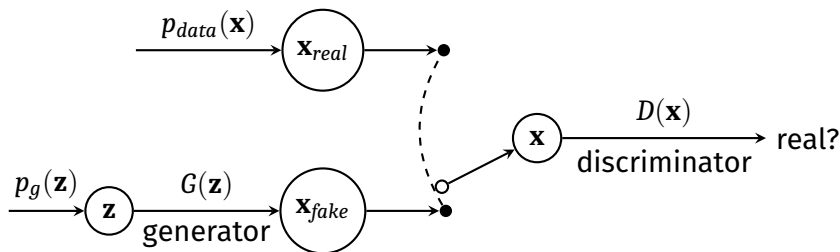
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Let  $\theta_g$  and  $\theta_d$  represent the parameters for the generator and discriminator neural networks, respectively.

# Generative Adversarial Net



Goodfellow, Ian J., Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. "Generative adversarial nets." *Advances in neural information processing systems* 27 (2014).

# Generative Adversarial Net

⚡ We have two neural network in a GAN

- The network  $G$ 's goal is to generate sample that look like they came from  $p_{data}(\mathbf{x})$ .
- The network  $D$ 's goal is to effectively discriminate between real samples from  $p_{data}(\mathbf{x})$  and fake samples from  $G$ .

⚡ The goals of  $G$  and  $D$  are different! We can view this scenario as a game between  $D$  and  $G$ .

*We train  $D$  to maximize the probability of assigning the correct label to both training examples and samples from  $G$ . We simultaneously train  $G$  to minimize  $\log(1 - D(G(z)))$ .*

# Formally Defining the Optimization Task

The optimization of  $\theta_g$  and  $\theta_d$  can be cast as a min-max task, which is given by

$$\min_G \max_D V(D, G) = \{\mathbb{E}_{\mathbf{x} \sim p_{data}} [\log(D(\mathbf{x}))] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))]\} \quad (1)$$

$V$  is called as value function.

From the original paper: alternate between  $k$  steps of optimizing  $D$  and one step of optimizing  $G$ .

In the beginning  $G$  is poor, so  $D$  will reject samples with a high confidence, and  $\log(1 - D(G(z)))$  saturates. So, rather than training  $G$  to minimize  $\log(1 - D(G(z)))$ ,  $G$  is trained to maximize  $\log D(G(z))$ .



# Formally Defining the Optimization Task

- ⚡ If  $\mathbf{x} \sim p_{data}(\mathbf{x})$  then the discriminator should maximize  $\log D(\mathbf{x})$
- ⚡ If  $\mathbf{z} \sim p_g(\mathbf{z})$  then the generator seeks to generate a sample that makes the discriminator think it is a normal sample.
- ⚡ We can use stochastic gradient descent and ascent to adjust the parameters of the network.

## Global Optimality of $p_g = p_{\text{data}}$

If we fix the generator  $G$ , the optimal discriminator  $D$  is

$$D_G^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})}$$

# Minimax Equation with Optimal Discriminator

For a fixed  $G$ , the minimax is

$$\begin{aligned}C(G) &= \max_D V(D, G) \\&= \mathbb{E}_{\mathbf{x} \sim p_{data}} [\log(D_G^*(\mathbf{x}))] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D_G^*(G(\mathbf{z})))] \\&= \mathbb{E}_{\mathbf{x} \sim p_{data}} [\log(D_G^*(\mathbf{x}))] + \mathbb{E}_{\mathbf{x} \sim p_g} [\log(1 - D_G^*(\mathbf{x}))] \\&= \mathbb{E}_{\mathbf{x} \sim p_{data}} \left[ \log \frac{p_{data}(\mathbf{x})}{p_{data}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[ \log \frac{p_g(\mathbf{x})}{p_{data}(\mathbf{x}) + p_g(\mathbf{x})} \right]\end{aligned}$$

Here, training objective of  $D$  can be interpreted as maximizing the log-likelihood for estimating the conditional probability  $P(Y = y|\mathbf{x})$  where  $Y$  is the random variable denoting where  $\mathbf{x}$  comes from  $p_{data}$  with  $y = 1$  or from  $p_g$  with  $y = 0$ .

# Formally Defining the Optimization Task

**Theorem 1.** The global minimum of the virtual training criterion  $C(G)$  is achieved if and only if  $p_g = p_{data}$ . At that point,  $C(G)$  achieves the value  $-\log(4)$ . This point is a Nash Equilibrium.

# Pseudo Code for Training a GAN

**for**  $t = 1, \dots, T$

**for**  $k = 1, \dots, K$

- Sample  $m$  noise data  $\mathbf{z}_1, \dots, \mathbf{z}_m$  from  $p_g(\mathbf{z})$
- Sample  $m$  data samples  $\mathbf{x}_1, \dots, \mathbf{x}_m$  from  $p_{data}(\mathbf{x})$
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \left\{ \frac{1}{m} \sum_{i=1}^m [\log(D(\mathbf{x}_i)) + \log(1 - D(G(\mathbf{z}_i)))] \right\} \quad (2)$$

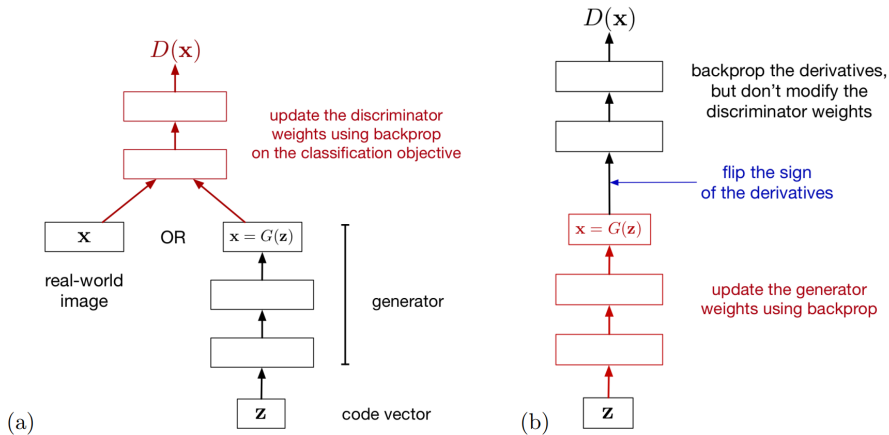
**end for**

- Sample  $m$  noise data  $\mathbf{z}_1, \dots, \mathbf{z}_m$  from  $p_g(\mathbf{z})$
- Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \left\{ \frac{1}{m} \sum_{i=1}^m \log(1 - D(G(\mathbf{z}_i))) \right\} \quad (3)$$

**end for**

# Formally Defining the Optimization Task



From Roger Grosse's GAN lecture notes.

# On Training GANs

## Challenge

One issue associated with training a GAN is the form of the loss. Observe above that the generator weights only get updated by the gradients of the second term:

$$\log(1 - D(G(\mathbf{z}))) \quad (4)$$

- ⚡ The problem with this is that if the generator sample is really bad, then the discriminator's prediction is close to zero, and since  $\log(1 - D)$  is very flat when  $D \approx 0$  there is not enough "signal" to move the generator weights meaningfully.
- ⚡ Increasing learning rates do not seem to help. This is called the *saturation problem* in GANs.
- ⚡ To fix this, while updating generator weights, it is common to heuristically replace the second term in the GAN loss with  $-\log D(G(\mathbf{z}))$

# On Training GANs

## Challenge

Stably training GANs was a challenge faced by the community for quite some time (and continues to be a challenge), and a common resolution is to use *Wasserstein GANs*.

- ⚡ The GAN loss is a form of distance between probability distributions (called the Jensen-Shannon divergence), and this can be generated to other distances.
- ⚡ A common alternative is the Earth-mover or Wasserstein distance, leading to a different type of GAN model called Wasserstein GAN. The WGAN was shown to improve stability of the optimization process



# GANs (in PyTorch)

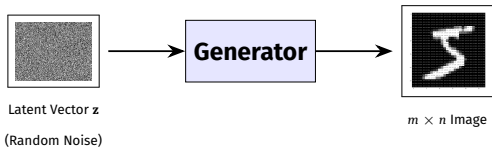
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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |

# Generators

## Conceptual Flow

The Generator takes random noise (latent vector  $z$ ) and generates an  $m \times n$  image.



## PyTorch Implementation

```
1 class Generator(nn.Module):
2     def __init__(self, latent_dim=100, height=28, width=28):
3         super(Generator, self).__init__()
4         self.latent_dim = latent_dim
5         self.height, self.width = height, width
6
7     self.model = nn.Sequential(
8         nn.Linear(latent_dim, 256),
9         nn.LeakyReLU(0.2),
10        nn.Linear(256, 512),
11        nn.LeakyReLU(0.2),
12        nn.Linear(512, 1024),
13        nn.LeakyReLU(0.2),
14        nn.Linear(1024, width*height),
15        nn.Tanh() # Output in [-1, 1]
16    )
17
18    def forward(self, z):
19        img = self.model(z)
20        # Reshape to [batch, channels, H, W]
21        return img.view(img.size(0), 1,
22                        self.height, self.width)
```

# Discriminators

## Conceptual Flow

The discriminator takes generated image and guess whether it is fake or real through binary classification.



## PyTorch Implementation

```
1 class Discriminator(nn.Module):
2     def __init__(self, height = 28, width = 28):
3         super(Discriminator, self).__init__()
4         self.height, self.width = height, width
5
6         self.model = nn.Sequential(
7             nn.Linear(self.height*self.width, 1024),
8             nn.LeakyReLU(0.2),
9             nn.Dropout(0.3), # Regularization
10            nn.Linear(1024, 512),
11            nn.LeakyReLU(0.2),
12            nn.Dropout(0.3),
13            nn.Linear(512, 256),
14            nn.LeakyReLU(0.2),
15            nn.Dropout(0.3),
16            nn.Linear(256, 1),
17            nn.Sigmoid() # Output probability [0, 1]
18        )
19
20    def forward(self, img):
21        img_flat = img.view(img.size(0), -1) # Flatten
22        validity = self.model(img_flat)
23        return validity
```

# Training Loss in GAN Implementation

Training loss is actually just Binary Cross Entropy loss effectively.

$$\mathcal{L}_{BCE}(y, \hat{y}) = -[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})]$$

**Case 1: Real Sample ( $y = 1$ )**

$$\mathcal{L}_{BCE} = -\log D(x)$$

**Case 2: Fake Sample ( $y = 0$ )**

$$\mathcal{L}_{BCE} = -\log(1 - D(G(z)))$$

Exactly the negative of the value function mentioned in the original paper.

We also noted earlier that GoodFellow recommended to maximize  $\log D(G(z))$  instead of minimizing  $\log(1 - D(G(z)))$ , because early training:  $D$  easily detects fakes  $\Rightarrow D(G(z)) \approx 0$

$$\frac{\partial}{\partial \theta} \log(1 - D(G(z))) \approx 0 \quad \textbf{(Vanishing gradients!)}$$

**Theory (Maximize)**

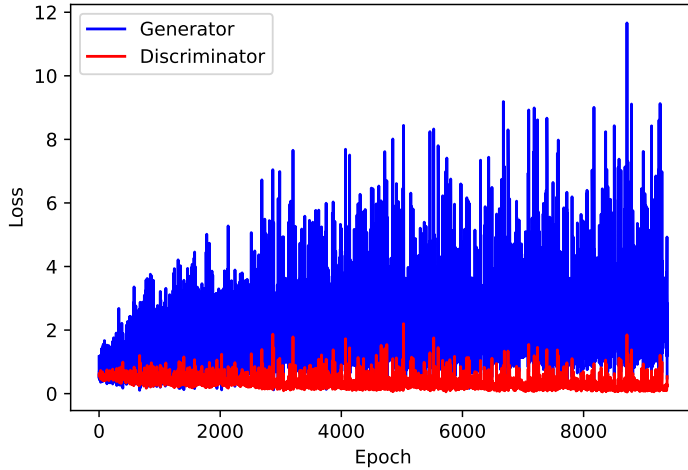
$$\begin{aligned} &\mathbb{E}[\log D(x)] \\ &\mathbb{E}[\log(1 - D(G(z)))] \end{aligned}$$

**Code (Minimize)**

$$\begin{aligned} &\text{criterion}(D(x), 1) \\ &\text{criterion}(D(G(z)), 0) \end{aligned}$$

**Implementation: Just flip the label!**

# GAN Loss

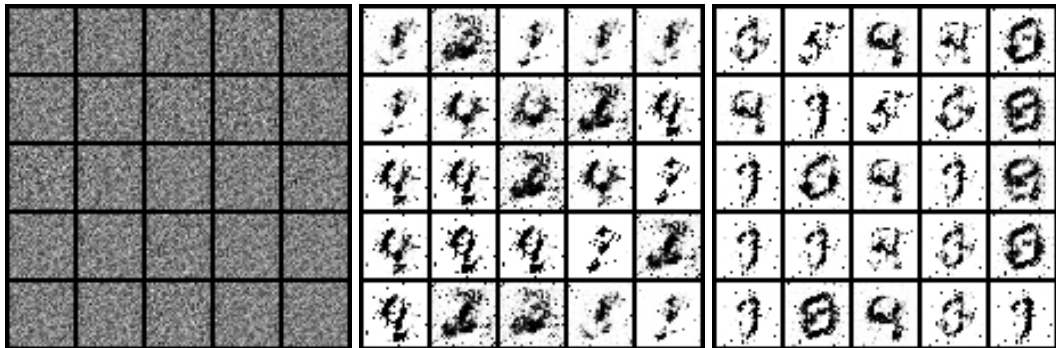


# Examples

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |

# GAN example



# GAN example: Sketch to Figure





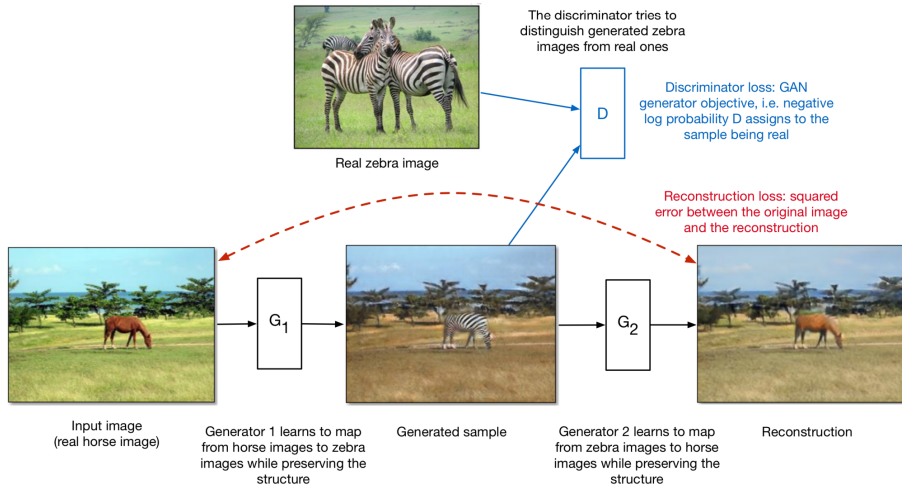
## GAN example: Cycle GANs

- ⚡ Image-to-image translation involves generating a new synthetic version of a given image with a specific modification, such as translating a summer landscape to winter

# GAN example: Cycle GANs

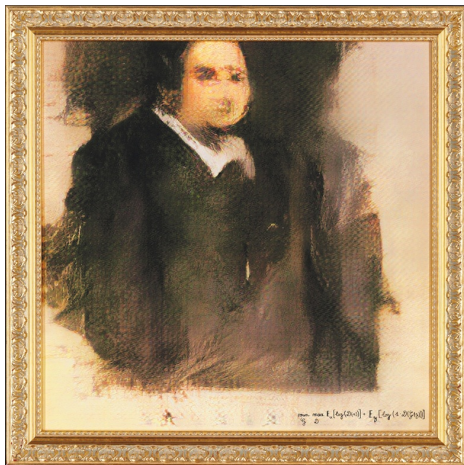


# GAN example: Cycle GANs



$$\text{Total loss} = \text{discriminator loss} + \text{reconstruction loss}$$

# First AI-Generated Portrait Ever Sold at Auction Shatters Expectations: \$432,500



# Cosmopolitan Meets DALL-E 2



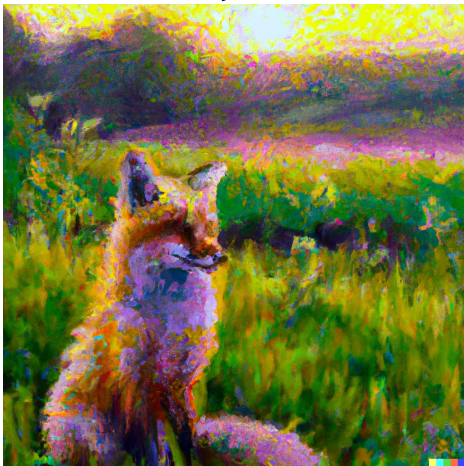
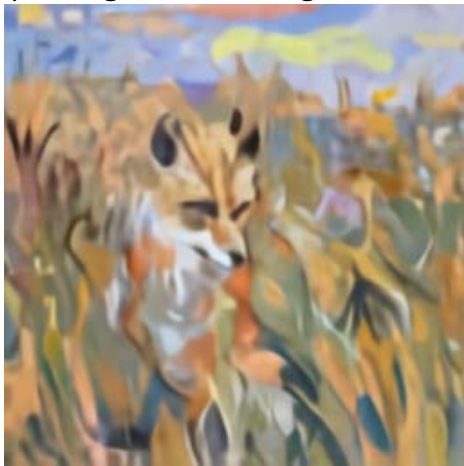
## DALL-E 2: Text-to-Image Synthesis

- ⚡ DALL-E 2 is a new AI system that can create realistic images and art from a description in natural language. DALL-E 2 can create original, realistic images and art from a text description. It can combine concepts, attributes, and styles.
- ⚡ DALL-E's uses multimodal version of GPT-3 with 12 billion parameters. The concept is that it swaps text for pixels and it is trained on trained on text-image pairs from the Internet. DALL-E 2 uses 3.5 billion parameters.

## DALL-E 2

“a painting of a fox sitting in a field at sunrise in the style of Claude Monet”

“a painting of a fox sitting in a field at sunrise in the style of Claude Monet”





# Challenges with GANs

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## Issues with GANs (Google's Comments)

- ⚡ Research has suggested that if your discriminator is too good, then generator training can fail due to vanishing gradients. In effect, an optimal discriminator doesn't provide enough information for the generator to make progress.
  - The Wasserstein loss is designed to prevent vanishing gradients even when you train the discriminator to optimality.
- ⚡ The generator may learn to produce a single output. The generator is always trying to find the one output that seems most plausible to the discriminator. This form of GAN failure is called *mode collapse*.
- ⚡ GANs frequently fail to converge, as discussed in the module on training.

# Advance Architecture in GAN

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# Deep Convolution GAN

Fractionally-Strided Convolutional Layer (or Transposed Convolution) is opposite of convolution operation.

In a convolution operation (for example, stride = 2), we obtain a downsampled (smaller) output of the larger input.

In fractionally-strided convolution, an upsampled (larger) output is obtained from a smaller input.

Consider  $X = \begin{bmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{bmatrix}$  and filter as  $K = \begin{bmatrix} k_{00} & k_{01} & k_{02} \\ k_{10} & k_{11} & k_{12} \\ k_{20} & k_{21} & k_{22} \end{bmatrix}$

with stride of 2, i.e. fractional stride of 1/2, we insert, zeros to between every other input

pixel. Hence,  $X = \begin{bmatrix} x_{00} & 0 & x_{01} & 0 \\ 0 & 0 & 0 & 0 \\ x_{10} & 0 & x_{11} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ . Then we apply regular convolution operation.

*Radford, Alec, Luke Metz, and Soumith Chintala. "Unsupervised representation learning with deep convolutional generative adversarial networks." arXiv preprint arXiv:1511.06434 (2015).*

# Deep Convolution GAN

**Fractionally-Strided Convolutional Layer** (or Transposed Convolution) is the opposite of the convolution operation.

## Key Concepts:

- ⚡ **Standard Convolution:** In a convolution operation (for example, stride = 2), we obtain a downsampled (smaller) output of the larger input.
- ⚡ **Fractional Stride:** In fractionally-strided convolution, an upsampled (larger) output is obtained from a smaller input.

Radford, Alec, Luke Metz, and Soumith Chintala. "Unsupervised representation learning with deep convolutional generative adversarial networks." arXiv preprint arXiv:1511.06434 (2015).

**Mathematical Example:** Consider  $X = \begin{bmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{bmatrix}$  and filter as  $K = \begin{bmatrix} k_{00} & k_{01} & k_{02} \\ k_{10} & k_{11} & k_{12} \\ k_{20} & k_{21} & k_{22} \end{bmatrix}$

We insert zeros between every other input pixel. Hence:

$$X = \begin{bmatrix} x_{00} & 0 & x_{01} & 0 \\ 0 & 0 & 0 & 0 \\ x_{10} & 0 & x_{11} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Then we apply regular convolution operation.

# Deep Convolution GAN

From the original paper:

- 1 Replace any pooling layers with strided convolutions (discriminator) and fractional-strided convolutions (generator).
- 2 Use batchnorm in both the generator and the discriminator.
- 3 Remove fully connected hidden layers for deeper architectures.
- 4 Use ReLU activation in generator for all layers except for the output, which uses Tanh.
- 5 Use LeakyReLU activation in the discriminator for all layers.

# References



Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, Yoshua Bengio (2013)

**Generative Adversarial Networks**

*Advances in Neural Information Processing Systems.*

# The End